Week 9: Bayesian regression

SOC6708 ADA

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1 Packages required

```
library(rstan)
library(tidyverse)
library(tidybayes)
library(janitor)
```

2 Reminder: what is a Gompertz model?

In 1825 Benjamin Gompertz, an actuary in London, proposed the following two-parameter model for mortality:

$$\mu(x) = \alpha e^{\beta x}$$

where $\mu(x)$ is the instantaneous mortality rate (hazard) at age x. This model assumes that mortality increases exponentially with age, which is a pretty good assumption for adult mortality (above age 40 or so, with the exception of older ages).

3 Data

For the following two examples we're going to using death and population counts by age and sex for Ontario, sourced from the Canadian Human Mortality Database project. We can read the data files in directly from the URLs:

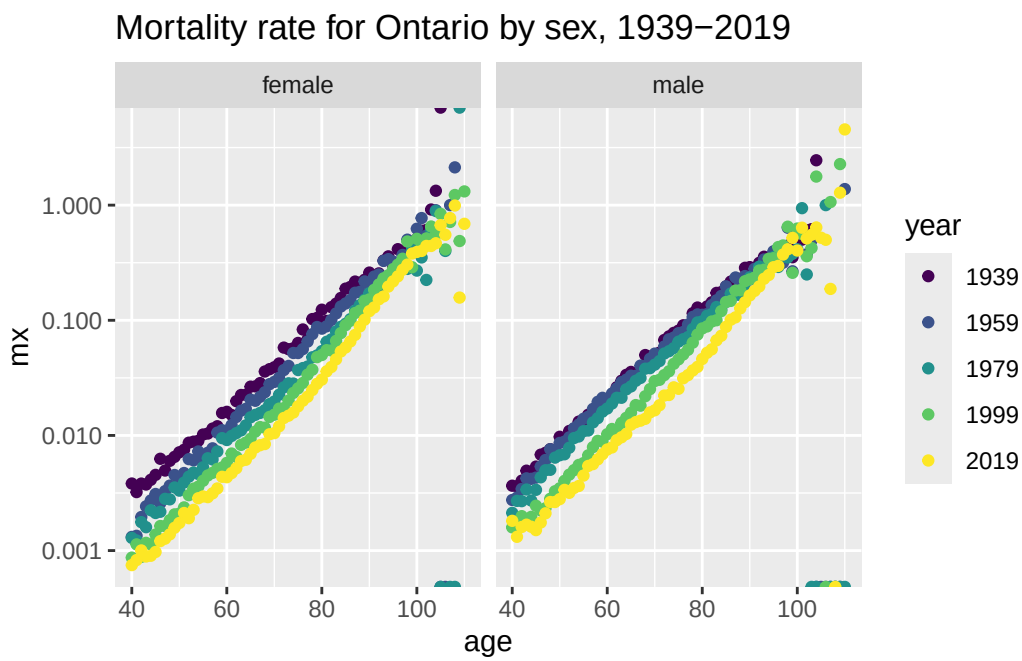
```
dd <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Deaths_1x1.txt", skip = 1)
dp <- read_table("https://www.prdh.umontreal.ca/BDLC/data/ont/Population.txt", skip = 1)
```

These data files are in 'wide' format. For our purposes it's going to be easier to work with in 'long format'. So let's do that and also clean some other stuff up:

```
dd <- dd |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "deaths") |>
  mutate(deaths = as.numeric(deaths), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

dp <- dp |>
  clean_names() |>
  pivot_longer(-(year:age), names_to = "sex", values_to = "pop") |>
  mutate(pop = as.numeric(pop), age = as.numeric(age)) |>
  mutate(age = ifelse(is.na(age), 110, age))

d <- dd |>
  left_join(dp) |>
  mutate(mx = deaths/pop,
         log_mx = log(mx))
```

Do some quick plots

```
d |>
  filter(age>39, year %in% seq(1939, 2019, by = 20), sex!= "total") |>
  ggplot(aes(age, mx, color = factor(year))) +
  geom_point() +
  facet_wrap(~sex)+
  labs(title = "Mortality rate for Ontario by sex, 1939-2019")+
  scale_y_log10()+
  scale_color_viridis_d(name = "year")
```



4 Base model

First let's fit a Gompertz model to males aged 40+ in 2019. Note that Gompertz models have the form

$$\mu(x) = \alpha e^{\beta x}$$

So we can write

$$\log \mu(x) = \log \alpha + \beta x$$

More notes:

- To make this fully Bayesian we need to specify the likelihood and priors. The full model here is

$$y_x \sim \text{Poisson}(P_x \cdot m_x)$$

$$\log m_x = \log \alpha + \beta x$$

$$\log \alpha \sim N(0, 10^2)$$

$$\beta \sim N(0, 0.1^2)$$

  where y_x is deaths at age x and P_x is population
- Could have used a normal likelihood (c.f. using `lm`) but nice to account for population size
- We are fitting not on age, but on a centered version (why?)

Now we need to get the data in the right format to read into Stan (this required a named list):

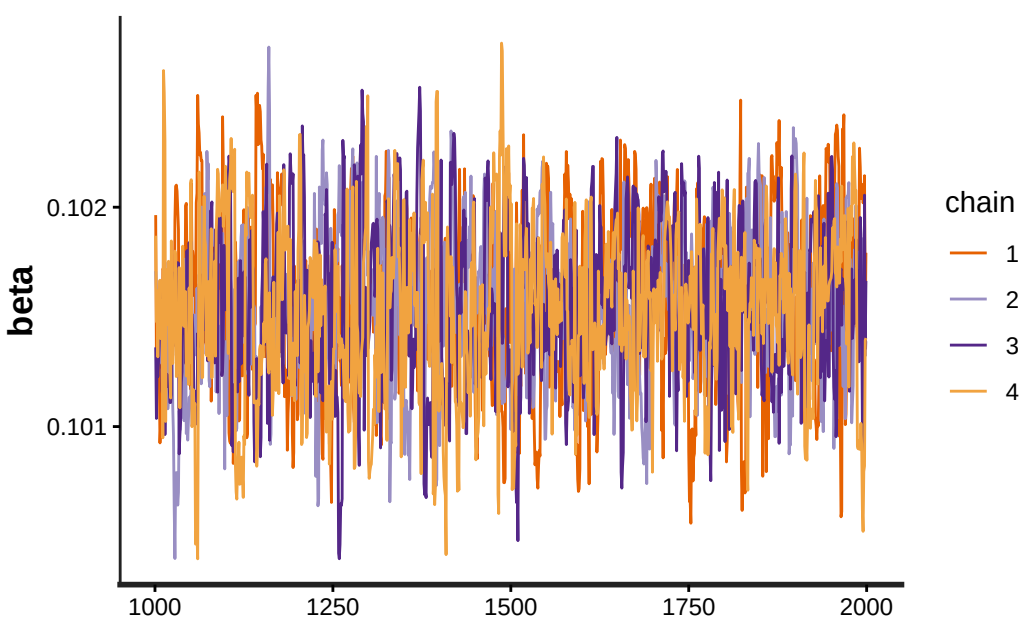
```
d_male_19 <- d |>
  filter(year==2019, sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

stan_data <- list(y = round(d_male_19$deaths),
                  pop = round(d_male_19$pop),
                  N = nrow(d_male_19),
                  age_c = d_male_19$age_c)
```

Run the model and look at some output:

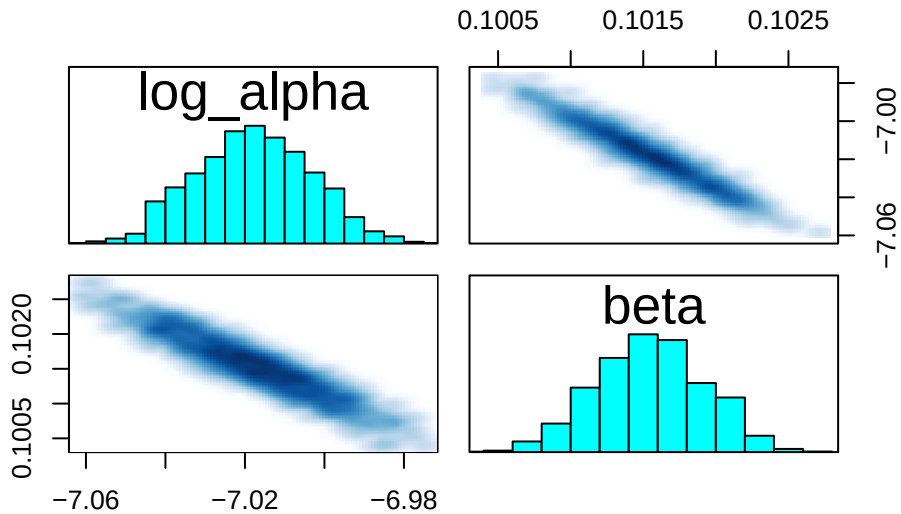
```
 mod <- stan(file = "C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/gomp.stan"
            data = stan_data,
            seed = 123)
mod
```

Some quick model checks, looking at the traceplot and pairwise densities

```
pars <- c("log_alpha", "beta")
traceplot(mod, pars = c("beta"))
```



```r
pairs(mod, pars = pars)
```



Chains are overlapping a lot; fluCtuating within the same range Off-diagonal plots show negative relationship

```r
temp <- rstan::extract(mod)
names(temp)
```

```
[1] "log_alpha" "beta"      "mu"        "lp__"
```

```r
temp$log_alpha
```

```
  [1] -7.043870 -7.019617 -7.002325 -7.030377 -7.034142 -7.004943 -7.023546
  [8] -7.007738 -7.007569 -7.018860 -7.021585 -7.001827 -7.003675 -7.016448
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```

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[3599] -7.010734 -7.016039 -7.025438 -7.024275 -7.018031 -7.002299 -7.009916
[3606] -7.027222 -7.009066 -6.997641 -7.029063 -6.985350 -7.019449 -6.982101
[3613] -7.024410 -7.028256 -7.000713 -7.024985 -7.046492 -7.029818 -7.040354
[3620] -7.028108 -6.986382 -7.043825 -7.013661 -7.034110 -7.004440 -7.005015
[3627] -7.033641 -7.002994 -7.040733 -7.012612 -7.007375 -7.007519 -7.037542
[3634] -7.024318 -7.021325 -7.017410 -6.999922 -7.005410 -6.985421 -7.015278
[3641] -7.042610 -7.028544 -7.005478 -7.004050 -7.027972 -7.000878 -7.011566
[3648] -6.985767 -7.033489 -7.029399 -7.015671 -7.007034 -7.003392 -7.028884
[3655] -7.015951 -7.019210 -6.999040 -6.998442 -6.995697 -7.028985 -7.011819
[3662] -7.009808 -7.005033 -7.006064 -7.019007 -6.995555 -7.021184 -7.024710
[3669] -7.035875 -7.000341 -7.009630 -7.030788 -7.031606 -7.020056 -7.022819
```

```
[3676] -7.005199 -7.028831 -7.004789 -7.021972 -7.006746 -6.999844 -7.023885
[3683] -7.000662 -7.016134 -7.015884 -7.029964 -7.028938 -7.026627 -7.008520
[3690] -7.004200 -7.016674 -6.987847 -7.023028 -7.024312 -7.025617 -7.028376
[3697] -7.011836 -7.029117 -7.045651 -6.990118 -7.041376 -6.989887 -7.026992
[3704] -7.044965 -7.023221 -7.004253 -6.998183 -7.002269 -7.027662 -7.003466
[3711] -7.020092 -7.017309 -7.032337 -7.011896 -7.002311 -7.012067 -7.007685
[3718] -7.022224 -7.036851 -7.025873 -7.015629 -7.022470 -7.014891 -7.037659
[3725] -7.025408 -7.016081 -7.020046 -7.024948 -7.014971 -7.006055 -7.014146
[3732] -7.034518 -7.010211 -7.010177 -6.995406 -7.040380 -7.021286 -7.010421
[3739] -7.020290 -7.041405 -7.013512 -6.998913 -7.020053 -7.010704 -7.039007
[3746] -6.999043 -7.015754 -7.024630 -7.017517 -7.033766 -7.056421 -7.013263
[3753] -7.022855 -7.030743 -7.012064 -7.019713 -7.021598 -7.020925 -6.998065
[3760] -7.042577 -7.017537 -7.011549 -7.005310 -7.033403 -7.005443 -7.019294
[3767] -6.997967 -7.020211 -7.021834 -7.006935 -7.005414 -6.997815 -7.028593
[3774] -7.037559 -7.021627 -7.040216 -7.053076 -7.012147 -7.015927 -7.021323
[3781] -7.009825 -7.013872 -7.017288 -7.017179 -7.029543 -7.029840 -7.018891
[3788] -7.006461 -7.013012 -7.025985 -7.003678 -6.995672 -7.019856 -7.028033
[3795] -7.030388 -7.008505 -7.013776 -7.046064 -7.032198 -7.027292 -7.019527
[3802] -6.998814 -6.992899 -7.010799 -7.022861 -7.021902 -7.038403 -6.997775
[3809] -7.025501 -7.012567 -7.017024 -7.017771 -7.009879 -6.986779 -7.011114
[3816] -6.995878 -7.028396 -7.014936 -7.014621 -7.012763 -7.017750 -7.029753
[3823] -7.029573 -7.014595 -7.019701 -7.041216 -6.989441 -6.999736 -7.013871
[3830] -7.025868 -6.981756 -7.009143 -7.057788 -7.034841 -7.034364 -7.030935
[3837] -7.044438 -7.016275 -7.002468 -7.017741 -7.028133 -7.023642 -7.014788
[3844] -7.027684 -7.026743 -6.995101 -7.003825 -7.014526 -7.021227 -7.003854
[3851] -7.017558 -7.013918 -7.015821 -7.013136 -7.010046 -6.991426 -7.042577
[3858] -7.008089 -7.043624 -7.012303 -7.016615 -7.009472 -7.016685 -7.018987
[3865] -7.004007 -7.002178 -7.026790 -6.989244 -7.005269 -6.978585 -7.031176
[3872] -7.021669 -6.990872 -7.006091 -7.029528 -7.002215 -7.030763 -7.025189
[3879] -7.030079 -7.014573 -7.004716 -7.020463 -7.019449 -7.015994 -6.987801
[3886] -7.026707 -7.013899 -7.037368 -7.035425 -7.009806 -7.015023 -7.028928
[3893] -7.017706 -7.032093 -7.015224 -7.002215 -7.002985 -7.045753 -7.017527
[3900] -7.008776 -6.992519 -7.039622 -7.024943 -7.025747 -6.993760 -7.032284
[3907] -7.022520 -7.015009 -7.005009 -6.998002 -7.018510 -7.019582 -7.034908
[3914] -7.010072 -6.996738 -7.021993 -6.997883 -7.028599 -7.018871 -7.026026
[3921] -7.006035 -7.023496 -7.029913 -7.021473 -7.024507 -7.027158 -7.034930
[3928] -7.016309 -7.029470 -7.013204 -7.022731 -7.036935 -7.025569 -7.027495
[3935] -7.024492 -7.004521 -7.035849 -7.018975 -6.987090 -7.003155 -7.000541
[3942] -7.029450 -7.007140 -7.028195 -6.999473 -7.037003 -7.012026 -7.014023
[3949] -7.024927 -7.042359 -7.030353 -6.994786 -7.023743 -7.025616 -7.002855
[3956] -7.019349 -7.014837 -7.031913 -7.008440 -7.032848 -7.024787 -7.013673
[3963] -7.030133 -7.021975 -7.016885 -7.058193 -7.035675 -7.009334 -7.010691
[3970] -7.007238 -7.033217 -7.017139 -7.027323 -7.025616 -6.994397 -7.026181
```

```
[3977] -7.018405 -7.028208 -7.016313 -7.006764 -7.015838 -6.997393 -7.017018
[3984] -7.028285 -7.006955 -7.026845 -7.023133 -7.045398 -7.000320 -7.034639
[3991] -7.006573 -7.004609 -7.036841 -6.990196 -7.020596 -7.020745 -7.002311
[3998] -7.030331 -7.009277 -6.993049
```

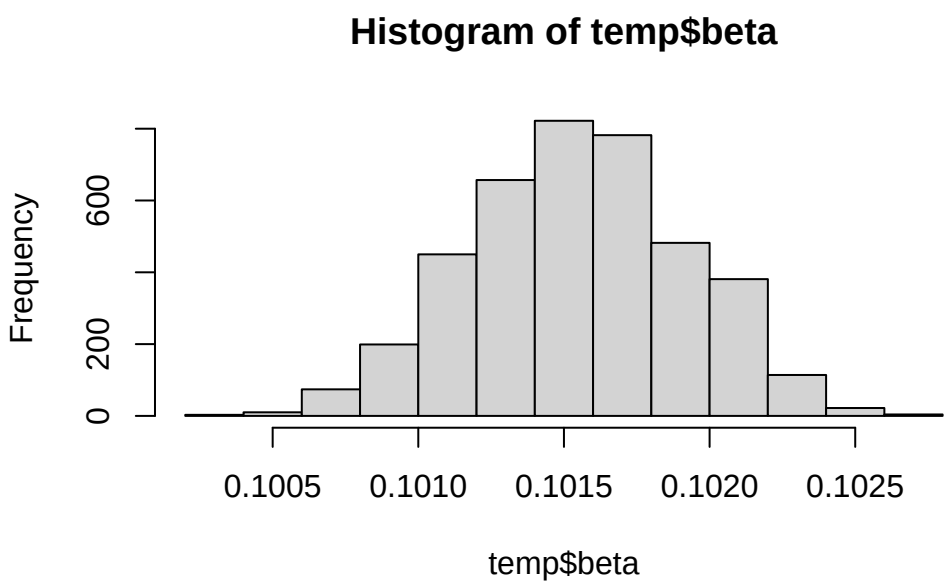
```r
median(temp$log_alpha)
```

```
[1] -7.017749
```

```r
median(temp$beta)
```

```
[1] 0.1015469
```

```r
hist(temp$beta)
```




```r
# 95% credible interval for beta
quantile(temp$beta, 0.025) # lower
```

```
     2.5%
0.1008201
```

```r
quantile(temp$beta, 0.975)  # upper
```

```
    97.5%
0.1022371
```

4.1 Extract the parameter estimates using `tidyverse`

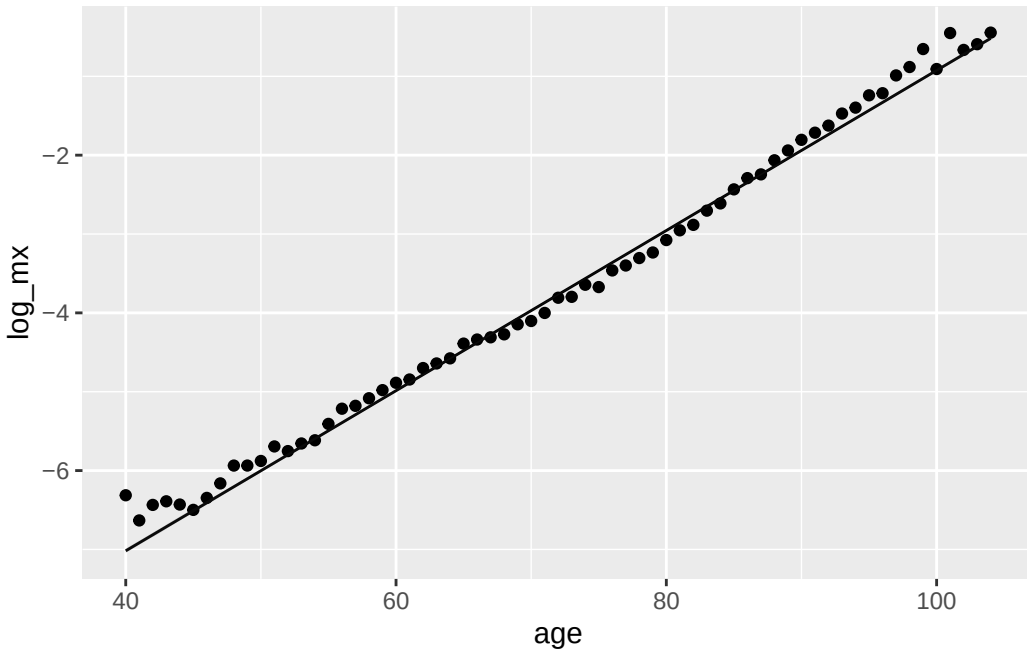
```r
mod |>
  gather_draws(log_alpha, beta) |>
  median_qi(.width = 0.99)
```

```
# A tibble: 2 x 7
  .variable .value .lower .upper .width .point .interval
  <chr>      <dbl>  <dbl>  <dbl>  <dbl> <chr>  <chr>
1 beta       0.102  0.101  0.102   0.99 median qi
2 log_alpha -7.02  -7.05  -6.98    0.99 median qi
```

4.2 Calculate and plot some results

Now we can combine the `tidybayes` syntax with `ggplot` to plot some results. For example, here's the data versus the estimates for the linear predictor:

```r
mod |>
  gather_draws(mu[x]) |>
  median_qi() |>
  mutate(age_c = x - 1) |>
  left_join(d_male_19) |>
  ggplot(aes(age, log_mx)) +
  geom_point() +
  geom_line(aes(age, .value)) +
  geom_ribbon(aes(x = age, ymin = .lower, ymax = .upper), alpha = 0.2)
```



Let's calculate the modal age at death (which for Gompertz mortality is a function of α and β, see here).

```
mod |>
  spread_draws(log_alpha, beta) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))) |>
  median_qi()
```

```
# A tibble: 1 x 12
  log_alpha log_alpha.lower log_alpha.upper  beta beta.lower beta.upper mode_age
      <dbl>           <dbl>           <dbl> <dbl>      <dbl>      <dbl>    <dbl>
1     -7.02           -7.04           -6.99 0.102      0.101      0.102     46.6
# i 5 more variables: mode_age.lower <dbl>, mode_age.upper <dbl>, .width <dbl>,
#   .point <chr>, .interval <chr>
```

Based on the posterior distribution, the modal age at death is very likely between about 46.5 and 46.7 years.

4.3 Question

Check you get similar results using `lm` or `glm`.

```r
summary(lm(log_mx ~ age_c, data = d |>
             filter(!is.infinite(log_mx)) |>
                      mutate(age_c = age-40)
                    ))
```

```
Call:
lm(formula = log_mx ~ age_c, data = mutate(filter(d, !is.infinite(log_mx)),
    age_c = age - 40))

Residuals:
    Min      1Q  Median      3Q     Max
-3.5022 -0.5513  0.0561  0.4002  6.5859

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -5.6637269  0.0052037 -1088.4   <2e-16 ***
age_c        0.0751362  0.0001534   489.9   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8721 on 33294 degrees of freedom
  (608 observations deleted due to missingness)
Multiple R-squared:  0.8782,    Adjusted R-squared:  0.8782
F-statistic: 2.4e+05 on 1 and 33294 DF,  p-value: < 2.2e-16
```

```r
d |> filter(!is.infinite(log_mx)) |>
              filter(pop>0) |>
                      mutate(age_c = age-40) |>
              mutate(deaths = as.integer(deaths),
                     pop = as.integer(pop)) |>
  filter(!is.infinite(deaths)) |>
  filter(!is.infinite(pop)) |>
  View()

summary(glm(deaths ~ age_c + offset(log(pop)),
                                    family = "poisson",data = d |> filter(!is.infinite(log_m
                      mutate(age_c = age-40) |>
              mutate(deaths = as.integer(deaths),
                     pop = as.integer(pop)) |>
                            filter(pop>0) |>
```

```r
  filter(!is.infinite(deaths)) |>
  filter(!is.infinite(pop))
        ))
```

```
Call:
glm(formula = deaths ~ age_c + offset(log(pop)), family = "poisson",
    data = filter(filter(filter(mutate(mutate(filter(d, !is.infinite(log_mx)),
        age_c = age - 40), deaths = as.integer(deaths), pop = as.integer(pop)),
        pop > 0), !is.infinite(deaths)), !is.infinite(pop)))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -5.816e+00  5.327e-04  -10917   <2e-16 ***
age_c        7.399e-02  1.611e-05    4593   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 36321057  on 33132  degrees of freedom
Residual deviance:  7505026  on 33131  degrees of freedom
AIC: 7730825

Number of Fisher Scoring iterations: 6
```

The glm() is the better classical comparison. The lm() is the least appropriate, as it assumes normal errors, and ignores population. Though the estimates for glm() are different from those of the Bayesian model, since the former is using ALL the years.

5 Model over time

Let's fit a slightly more complicated model, for multiple years, where the coefficients themselves are modeled as a random walk over time, i.e.

$$\beta_t \sim N(\beta_{t-1}, \sigma_\beta^2)$$

That is, a different set of Gompertz parameters are fit to every year, but we are assuming that the values in the current year are related to those is the previous year. This is a form

of dynamic linear regression. We need to put priors on the first time point, and also on the variance terms:

$$\log \alpha_1 \sim N(-6, 1)$$
$$\beta_1 \sim N(0.1, 0.1^2)$$
$$\sigma^\alpha, \sigma_\beta \sim N^+(0, 1)$$

Note that the likelihood and model on mortality rates are as before, we just have an additional subscript for time:

$$y_{x,t} \sim \text{Poisson}(P_{x,t} \cdot m_{x,t})$$
$$\log m_{x,t} = \log \alpha_t + \beta_t x$$

Now to fit the model. First get the data in the right format:

```
years <- 1989:2019
d_male <- d |> filter(year>=years[1], sex == "male", age>39, age<105) |>
  mutate(age_c = age - 40)

y <- d_male |>
  select(age, year, deaths) |>
  pivot_wider(names_from = "year", values_from = "deaths") |>
  select(-age) |>
  as.matrix()

pop <- d_male |>
  select(age, year, pop) |>
  pivot_wider(names_from = "year", values_from = "pop") |>
  select(-age) |>
  as.matrix()

stan_data <- list(y = y,
                  pop = pop,
                  N = nrow(d_male_19),
                  age_c = d_male_19$age_c,
                  T = ncol(y))
```

Now fit the model (note: takes a while):

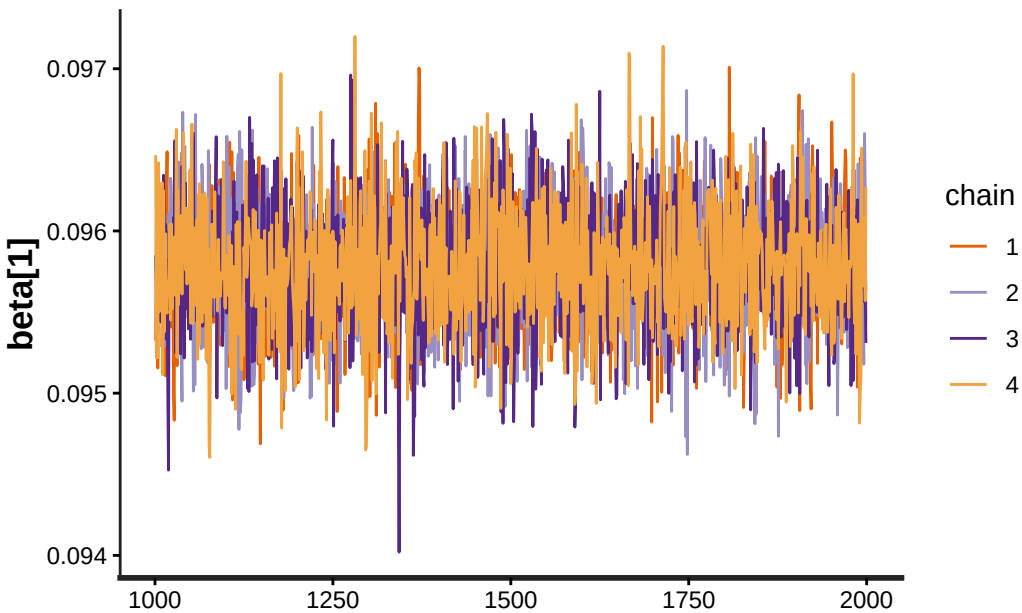
```
mod <- stan(file = "C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/gomp_time.st
            data = stan_data,
            seed = 852,
            refresh = 0)
```

```r
summary(mod)$summary[paste0("beta[", 1:(length(years)), "]"),]
```

```
               mean      se_mean           sd       2.5%        25%        50%
beta[1]  0.09578068 5.363219e-06 0.0003783758 0.09504679 0.09552250 0.09577536
beta[2]  0.09561034 4.960644e-06 0.0003459685 0.09494785 0.09537471 0.09560521
beta[3]  0.09666297 4.708501e-06 0.0003431123 0.09598514 0.09642029 0.09667122
beta[4]  0.09686813 4.816000e-06 0.0003457879 0.09619710 0.09663493 0.09686707
beta[5]  0.09707234 4.376008e-06 0.0003262593 0.09644113 0.09685285 0.09707489
beta[6]  0.09803360 4.560222e-06 0.0003315150 0.09737665 0.09780917 0.09803788
beta[7]  0.09872462 4.695849e-06 0.0003316200 0.09807991 0.09850131 0.09872110
beta[8]  0.09980847 4.998588e-06 0.0003317676 0.09914864 0.09958707 0.09980992
beta[9]  0.10070718 5.258703e-06 0.0003280055 0.10007723 0.10049526 0.10070063
beta[10] 0.10202266 4.901457e-06 0.0003364796 0.10136207 0.10180217 0.10201512
beta[11] 0.10279902 5.061565e-06 0.0003499438 0.10211529 0.10256646 0.10279591
beta[12] 0.10217902 4.907916e-06 0.0003384713 0.10153387 0.10194453 0.10217377
beta[13] 0.10153780 5.039963e-06 0.0003383654 0.10087959 0.10130364 0.10154076
beta[14] 0.10108711 4.745604e-06 0.0003361985 0.10041631 0.10085646 0.10107785
beta[15] 0.10135032 4.225887e-06 0.0003273955 0.10071613 0.10112695 0.10135628
beta[16] 0.10126576 4.398339e-06 0.0003264185 0.10063634 0.10104931 0.10126572
beta[17] 0.10169408 4.426359e-06 0.0003185674 0.10106705 0.10148576 0.10169292
beta[18] 0.10160828 4.175851e-06 0.0003166717 0.10098836 0.10138999 0.10160936
beta[19] 0.10157930 4.224511e-06 0.0003165494 0.10094866 0.10137229 0.10157909
beta[20] 0.10206846 4.051533e-06 0.0003145517 0.10144983 0.10186145 0.10206185
beta[21] 0.10203623 4.297264e-06 0.0003137067 0.10140839 0.10182617 0.10203893
beta[22] 0.10196988 4.264983e-06 0.0003089092 0.10137000 0.10176166 0.10197309
beta[23] 0.10201339 3.846781e-06 0.0003164922 0.10139393 0.10179551 0.10201901
beta[24] 0.10191524 4.073517e-06 0.0003090649 0.10131724 0.10170509 0.10191302
beta[25] 0.10226099 4.009263e-06 0.0003048083 0.10166336 0.10205160 0.10226381
beta[26] 0.10247620 4.103402e-06 0.0003025061 0.10187603 0.10226742 0.10248706
beta[27] 0.10239791 3.847264e-06 0.0003017187 0.10183950 0.10219088 0.10239227
beta[28] 0.10160847 4.083244e-06 0.0003167341 0.10099865 0.10138563 0.10160841
beta[29] 0.10211686 3.538670e-06 0.0002958289 0.10153745 0.10192180 0.10211978
beta[30] 0.10173831 3.901306e-06 0.0002921303 0.10117573 0.10154511 0.10173906
beta[31] 0.10119273 4.252874e-06 0.0002999170 0.10059951 0.10099049 0.10119281
                75%      97.5%    n_eff      Rhat
beta[1]  0.09603727 0.09651145 4977.321 0.9993266
beta[2]  0.09584456 0.09629882 4864.037 0.9996164
beta[3]  0.09689943 0.09733595 5310.153 0.9994338
beta[4]  0.09710114 0.09754820 5155.211 0.9997615
beta[5]  0.09729173 0.09771639 5558.653 0.9993781
beta[6]  0.09825685 0.09866246 5284.870 0.9999482
beta[7]  0.09894238 0.09939329 4987.158 1.0000567
```

```
beta[8]  0.10003036 0.10045741 4405.276 0.9998384
beta[9]  0.10092592 0.10135428 3890.495 1.0006275
beta[10] 0.10224933 0.10269864 4712.671 1.0014108
beta[11] 0.10303216 0.10348985 4779.990 1.0005136
beta[12] 0.10241013 0.10284201 4756.082 0.9999249
beta[13] 0.10177175 0.10219221 4507.307 0.9998951
beta[14] 0.10131125 0.10174910 5018.899 0.9996859
beta[15] 0.10156697 0.10198995 6002.191 0.9993681
beta[16] 0.10148229 0.10191665 5507.721 0.9994007
beta[17] 0.10190502 0.10232678 5179.756 0.9995239
beta[18] 0.10182258 0.10223234 5750.804 0.9997251
beta[19] 0.10178559 0.10220183 5614.744 0.9995804
beta[20] 0.10227724 0.10268518 6027.613 0.9991950
beta[21] 0.10224250 0.10264307 5329.219 1.0005162
beta[22] 0.10218201 0.10257785 5245.983 0.9995895
beta[23] 0.10223000 0.10263440 6769.101 0.9995689
beta[24] 0.10212343 0.10251852 5756.526 0.9995712
beta[25] 0.10246382 0.10287120 5779.957 0.9993357
beta[26] 0.10267576 0.10306569 5434.759 0.9997178
beta[27] 0.10260790 0.10299947 6150.358 0.9993327
beta[28] 0.10182642 0.10221640 6016.984 0.9991852
beta[29] 0.10230853 0.10269203 6988.775 0.9993631
beta[30] 0.10193284 0.10231892 5607.035 1.0000068
beta[31] 0.10139041 0.10177555 4973.214 0.9993773
```

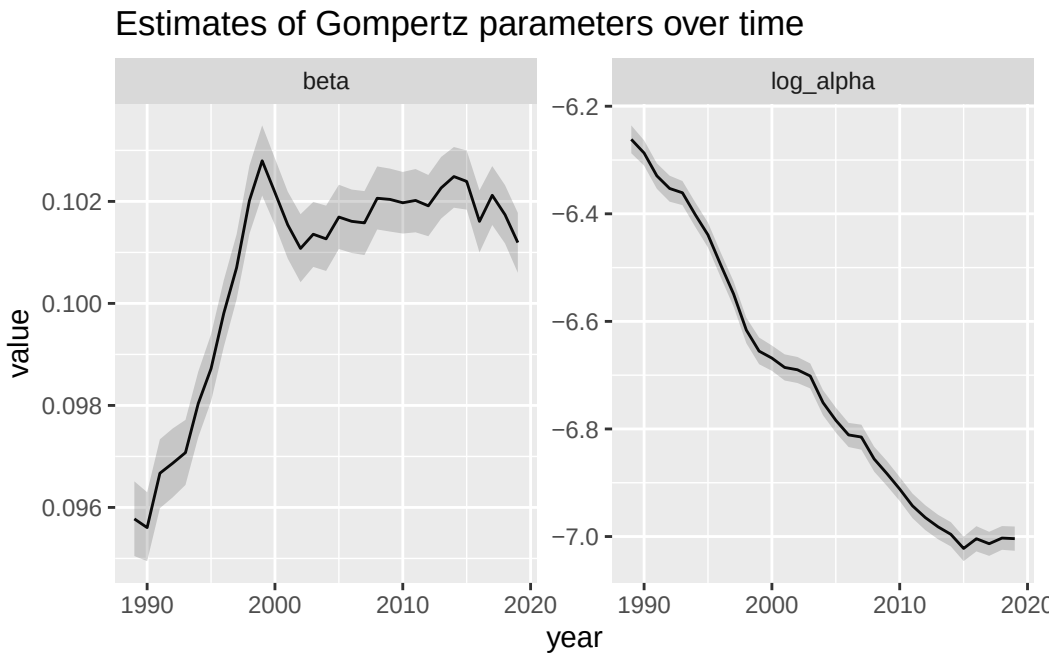
```r
traceplot(mod, pars = c("beta[1]"))
```



5.1 Plot parameter estimates over time

Now we can plot the parameter estimates over time:

```
mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, .value)) + geom_line() +
  facet_wrap(~.variable, scales = "free_y") +
  geom_ribbon(aes(ymin = .lower, ymax = .upper), alpha = 0.2)+
  labs(y = "value", title = "Estimates of Gompertz parameters over time")
```



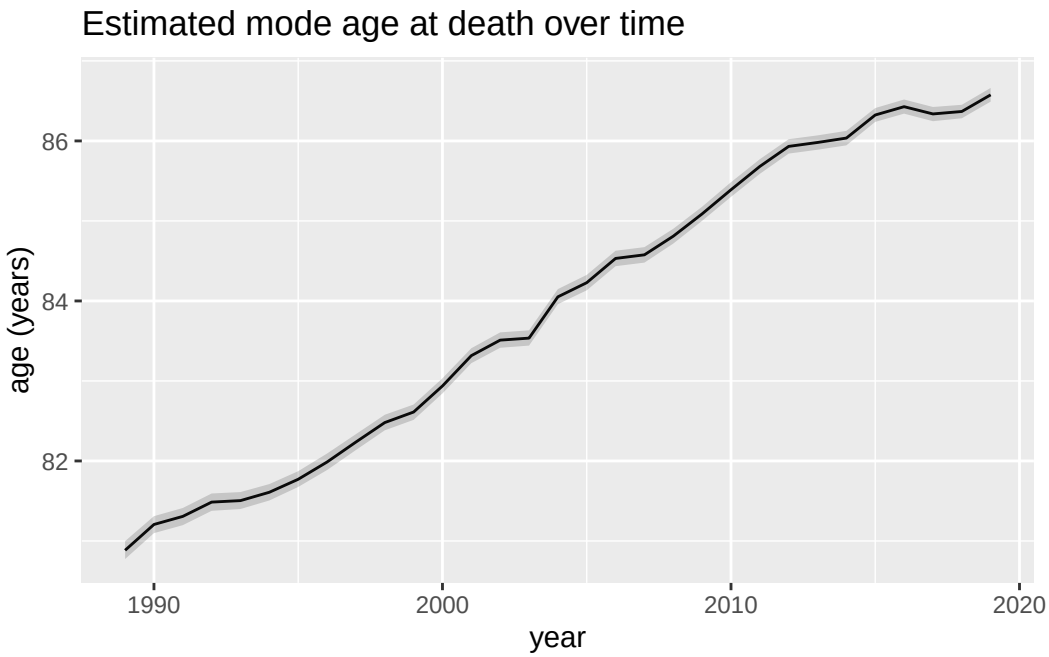
, which represents how quickly mortality increases with age, increases sharply from ~1990 to ~2000, and the it then levels off / stabilizes

log( ), baseline mortality level, has been going down

Mortality levels are going down, and mortality is increasing more sharply with age

…and also the mode age over time:

```
mod |>
  spread_draws(log_alpha[i], beta[i]) |>
  mutate(mode_age = 1/beta*log(beta/exp(log_alpha))+40) |>
  median_qi() |>
  mutate(year = years[i]) |>
  ggplot(aes(year, mode_age)) + geom_line() +
  geom_ribbon(aes(ymin = mode_age.lower, ymax = mode_age.upper), alpha = 0.2) +
  labs(title = "Estimated mode age at death over time", y = "age (years)")
```



5.2 Question

Can you forecast mortality rates with this model? If so, how? What are the assumptions behind the forecasts?

Yes, we can forecast mortality rates using this model by projecting the Gompertz parameters (and) forward in time. Future values of the parameters are simulated based on their previous values, and these are then used to generate age-specific mortality rates.

The assumptions behind the forecasts would be that changes in mortality follow a smooth random walk process, that past trends continue into the future, that the Gompertz functional form remains appropriate, and that no major structural changes occur.

6 Partial data observed

Now we're going to switch gears a bit and have a look at the situation where we have mortality rates for five geographic areas, but in one area we only have partial information. This is based on simulated data (you can have a look to see how I generated it based on the `simulated_data.R` script).

For 4 areas, we have deaths and population counts from ages 40 up to 60:

```r
df <- read_rds("C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/sim.rds")
df
```

```
# A tibble: 80 x 4
# Groups:   area [4]
     age  area deaths   pop
   <int> <int>  <int> <dbl>
 1    40     1      1  5000
 2    41     1      1  5000
 3    42     1      1  5000
 4    43     1      2  5000
 5    44     1      1  5000
 6    45     1      2  5000
 7    46     1      0  5000
 8    47     1      3  5000
 9    48     1      5  5000
10    49     1      3  5000
# i 70 more rows
```

For one region, we just have ${}_{20}q_{40}$, that is, the probability of dying between ages 40 and 60:

```r
q40 <- read_rds("C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/q40.rds")
q40
```

```
[1] 0.07262754
```

6.1 Relating ${}_{20}q_{40}$ to the Gompertz model

We want to fit a Gompertz model to each of the five areas (even the one with just a summary indicator). How to do this? Well, if we tell Stan how ${}_{20}q_{40}$ relates to μ_x, then the model has at least some information to estimate α and β. In particular, for each area, we're assuming:

$$ {}_1p_x = e^{-\mu_x} $$

where $\mu_x = \alpha e^{\beta x}$ and

$$ {}_{20}q_{40} = 1 - \prod_{x=40}^{60} {}_1p_x $$

Check the Stan file `gomp_partial.stan` to see this translated into code.

6.2 Fit the model

Get the data in the right format. Note that everything is a matrix now because we have more than one area:

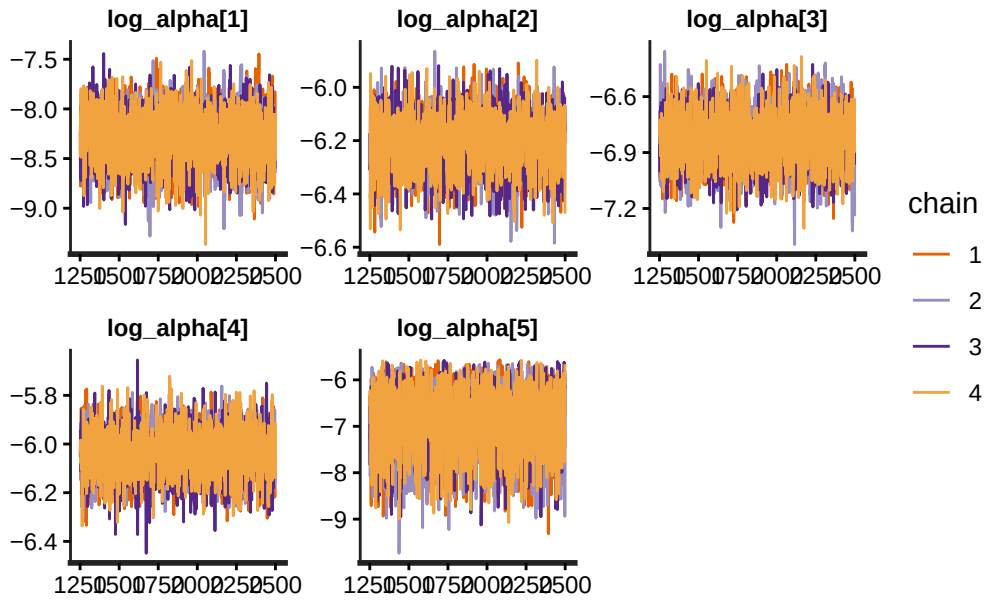
```
y <- as.matrix(df |>
  pivot_wider(names_from = "area", values_from = "deaths") |>
  select(-age, -pop))
pop <- as.matrix(df |>
  select(-deaths) |>
  pivot_wider(names_from = "area", values_from = "pop") |>
  select(-age))

stan_data <- list(N = length(40:59),
                  M = 4,
                  K = 5,
                  y = y,
                  pop = pop,
                  z = q40,
                  age_c = (40:59)-40)
```

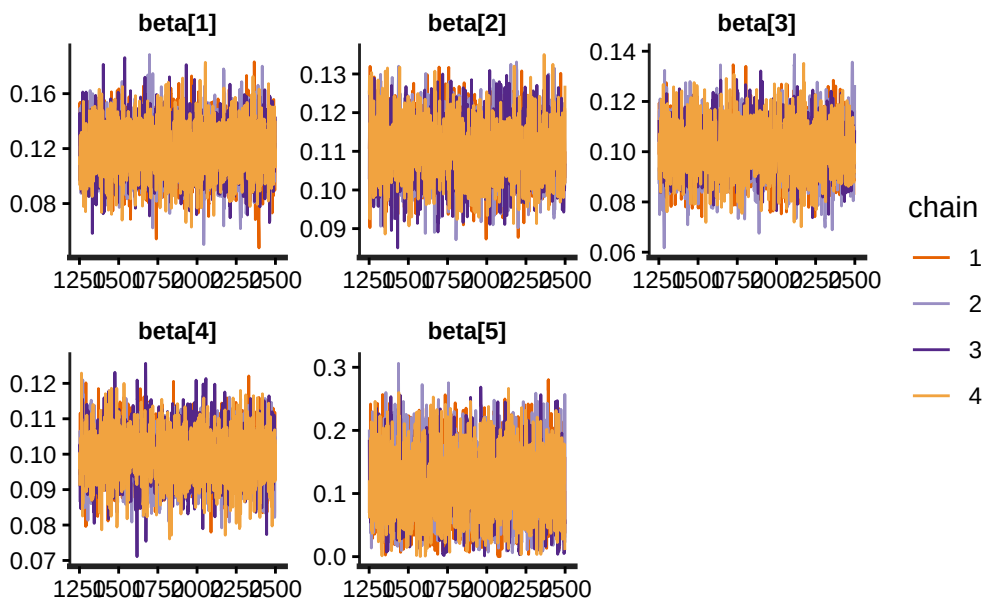
Fit the model:

```
mod <- stan(file = "C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/gomp_partial
            data = stan_data,
            seed = 15,
            control = list(adapt_delta = 0.96),
            iter = 2500,
            refresh = 0)
```

```
traceplot(mod, c("log_alpha"))
```

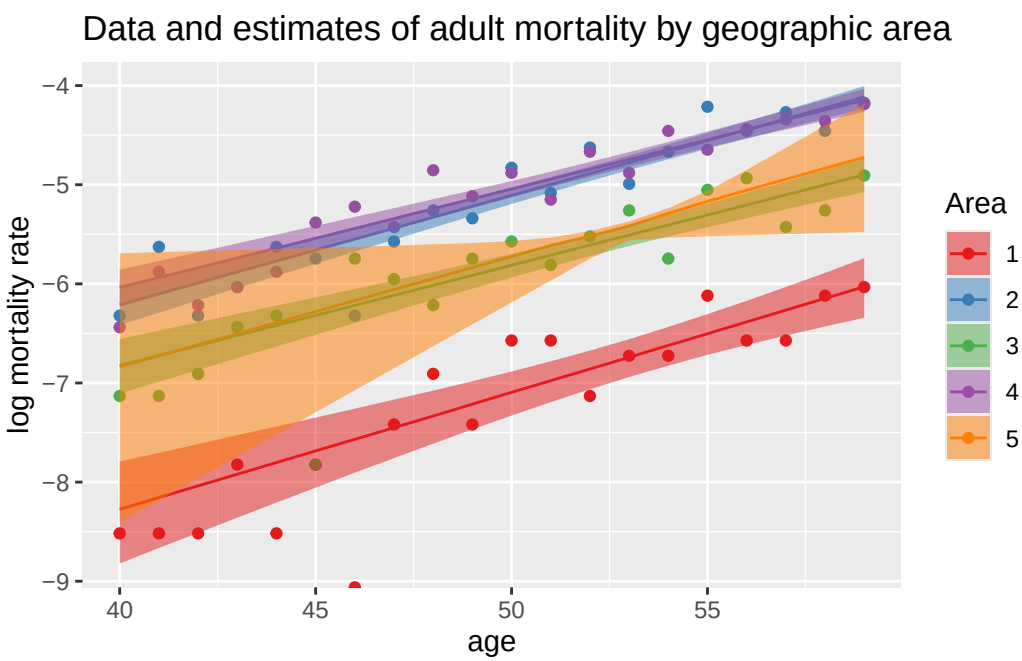



```
traceplot(mod, c("beta"))
```



Let's plot the fitted lines (with uncertainty) with the observed data for each region. Notice the difference in uncertainty around area 5!

```
mod |>
  gather_draws(mu[i,j]) |>
  median_qi() |>
  mutate(age = i-1+40) |>
  rename(area = j) |>
  left_join(df) |>
  mutate(log_mx = log(deaths/pop)) |>
  ggplot(aes(age, log_mx))+
  geom_point(aes(color = factor(area)))+
  geom_line(aes(age, .value, color = factor(area)))+
  geom_ribbon(aes(age, ymin = .lower, ymax = .upper, fill = factor(area)), alpha = 0.5)+
  scale_color_brewer(name = "Area", palette = "Set1")+
  scale_fill_brewer(name = "Area", palette = "Set1")+
  labs(title = "Data and estimates of adult mortality by geographic area", y = "log mortalit
```



Line 5: Fewer / noisier data points; Very wide uncertainty band

6.3 Compare parameter estimates against the truth

Here's what we estimated:

```r
mod |>
  gather_draws(log_alpha[i], beta[i]) |>
  median_qi()
```

```
# A tibble: 10 x 8
       i .variable  .value  .lower .upper .width .point .interval
   <int> <chr>       <dbl>   <dbl>  <dbl>  <dbl> <chr>  <chr>
 1     1 beta       0.118   0.0827  0.156   0.95 median qi
 2     1 log_alpha -8.27   -8.82   -7.79    0.95 median qi
 3     2 beta       0.110   0.0961  0.125   0.95 median qi
 4     2 log_alpha -6.21   -6.42   -6.02    0.95 median qi
 5     3 beta       0.101   0.0813  0.121   0.95 median qi
 6     3 log_alpha -6.82   -7.11   -6.56    0.95 median qi
 7     4 beta       0.0995  0.0859  0.113   0.95 median qi
 8     4 log_alpha -6.04   -6.23   -5.86    0.95 median qi
 9     5 beta       0.111   0.0109  0.222   0.95 median qi
10     5 log_alpha -6.84   -8.40   -5.69    0.95 median qi
```

And here's the truth (the values underlying the simulation):

```r
read_rds("C:/Users/hasan/OneDrive/Desktop/SOC PhD Stuff 2025, 2026/ADA/true_params.rds")
```

```
# A tibble: 5 x 3
    are log_alpha   beta
  <int>     <dbl>  <dbl>
1     1     -7.98 0.0985
2     2     -6.38 0.125
3     3     -6.88 0.110
4     4     -5.93 0.0924
5     5     -7.48 0.154
```

The model seems to work well. For all 5 areas, true values fall inside credible intervals